

THARUN KUMAR

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SUMMARY

ML and data engineer with 3 years building and operating production ML systems end to end, from feature pipelines through deployment, serving, and monitoring. Delivered demand and sales forecasting into planning workflows across retail and financial services, and owned the orchestration, release, and observability layers that keep those models running in production.

SKILLS

Languages: Python, SQL, R

ML & Forecasting: Time series forecasting, scikit-learn, TensorFlow, Transformers/BERT, NLTK, recommendation systems, anomaly detection, A/B testing

MLOps & Serving: MLflow, Kubeflow, SageMaker, Vertex AI, model registry and versioning, Docker, Kubernetes

Orchestration & CI/CD: Apache Airflow, Cloud Composer, GitHub Actions, Jenkins, Git, code review, technical documentation

Data: BigQuery, Snowflake, PostgreSQL, MySQL, SQL Server, Cassandra, S3, RDS, PySpark, Trino/Presto

Cloud: Google Cloud Platform, AWS, Azure

Analytics & BI: Tableau, Power BI, Alteryx, Matplotlib, Seaborn

EXPERIENCE

Tata Consultancy Services

ML Engineer

May 2023 – Jul 2024

- Owned ML delivery end to end for demand forecasting, credit risk, fraud, and segmentation, running packaging, registration, versioning, deployment, and serving on SageMaker, Kubeflow, and MLflow to cut release time by 45%.
- Built batch and near-real-time inference services on Docker and Kubernetes, tuned to hold latency and reliability targets at lower serving cost.
- Engineered ingestion, transformation, and feature pipelines across S3, RDS, and Snowflake, turning 8M+ records monthly into training and inference ready datasets.
- Orchestrated ML and data pipelines in Airflow with dependencies, retries, backfills, and SLA alerting, plus CI/CD promotion so releases shipped without manual steps.
- Implemented monitoring for model performance, data quality, and drift, alerting on threshold breaches to trigger retraining, and ran root cause analysis on failed runs and degraded model output.
- Automated data-quality validation (schema, null-rate, range, referential) that cut downstream inconsistencies by 40% across curated and star/snowflake models.

Data Scientist

Feb 2022 – Apr 2023

- Improved forecast accuracy by 20% with time series models for demand and sales planning, using seasonality and trend decomposition across product and location hierarchies.
- Established rolling-origin backtesting and a consistent accuracy metric, validating forecasts on held-out horizons before promotion.
- Built reusable feature-engineering and preprocessing components in Python, SQL, and PySpark, reused across forecasting and scoring workloads.
- Productionized models and scheduled scoring jobs through iterative Agile delivery, shortening the path from trained model to production scoring.

Data Engineer

Aug 2021 – Jan 2022

- Built and maintained ETL and ingestion pipelines in Python, SQL, and Alteryx, processing 3M+ records weekly.
- Improved query performance by 35% for analytics and training workloads through SQL and access-pattern optimization across PostgreSQL, SQL Server, and Cassandra.
- Built internal APIs exposing curated datasets to analytics and ML consumers, reducing duplicated transformation logic.
- Documented ETL processes, data contracts, and pipeline runbooks that kept analytics datasets on schedule.

PROJECTS

Cloud-Based Large-Scale Data Analysis | Google BigQuery (GCP)

May 2025

- Built an ELT workflow in BigQuery using advanced SQL for large-scale aggregation, trend analysis, and pattern identification.
- Published stakeholder-ready dashboards over the curated layer, making the analysis repeatable across business units.

Relational Database Design and SQL Development

Dec 2025

- Designed normalized schemas with ER diagrams for data integrity and referential consistency, and implemented complex SQL for retrieval and reporting.
- Modeled star and snowflake structures supporting BI consumption and feature-ready datasets.

EDUCATION

University of North Texas

Aug 2024 – May 2026

Master of Science, Information Systems

CERTIFICATIONS

Data Analysis with Python, IBM

Jan 2026

Google Data Foundations & Data Everywhere, Google

Oct 2025

AI Fluency and Claude 101, Anthropic

Jun 2026